

Research Notes on Bias variance tradeoff

Bias variance tradeoff

$$\text{MSE} = \text{Bias}^2 + \text{Variance} + \sigma_\epsilon^2$$

>> Section 1

$$\begin{aligned} E \left[(f(x) - f^{\wedge}(x))^2 \right] &= E \left[(f(x) - E[f^{\wedge}(x)] + E[f^{\wedge}(x)] - f^{\wedge}(x))^2 \right] \\ &= E \left[(f(x) - E[f^{\wedge}(x)])^2 \right] + E \left[(E[f^{\wedge}(x)] - f^{\wedge}(x))^2 \right] + 2 E \left[(f(x) - E[f^{\wedge}(x)])(E[f^{\wedge}(x)] - f^{\wedge}(x)) \right] \end{aligned}$$
$$\begin{aligned} E \left[(f(x) - E[f^{\wedge}(x)])^2 \right] &= E \left[(f(x) - \hat{f}(x))^2 \right] \\ E \left[(E[f^{\wedge}(x)] - f^{\wedge}(x))^2 \right] &= E \left[(E[f^{\wedge}(x)] - \hat{f}(x))^2 \right] \\ E \left[(f(x) - E[f^{\wedge}(x)])(E[f^{\wedge}(x)] - f^{\wedge}(x)) \right] &= 0 \end{aligned}$$

>> Section 2

In human learning While widely discussed in the context of machine learning, the bias–variance dilemma has been examined in the context of human cognition, most notably by Gerd Gigerenzer and co-workers in the context of learned heuristics. They have argued (see references below) that the human brain resolves the dilemma in the case of the typically sparse, poorly-characterized training-sets provided by experience by adopting high-bias/low variance heuristics. This reflects the fact that a zero-bias approach has poor generalizability to new situations, and also unreasonably presumes precise knowledge of the true state of the world. The resulting heuristics are relatively simple, but produce better inferences in a wider variety of situations. Geman et al. argue that the bias–variance dilemma implies that abilities such as generic object recognition cannot be learned from scratch, but require a certain degree of "hard wiring" that is later tuned by experience. This is because model-free approaches to inference require impractically large training sets if they are to avoid high variance.

>> Section 3

$$E \left[(y - \hat{f}(x; D))^2 \right] = (\text{Bias } D)^2 + \text{Var } D + \sigma^2$$

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>> Section 4

Approaches Dimensionality reduction and feature selection can decrease variance by simplifying models. Similarly, a larger training set tends to decrease variance. Adding features (predictors) tends to decrease bias, at the expense of introducing additional variance. Learning algorithms typically have some tunable parameters that control bias and variance; for example,

>> Section 5

$$E \left[(f(x) - \hat{f}(x))^2 \right] = E \left[(f(x) - E[f(x)] + E[f(x)] - \hat{f}(x))^2 \right]$$
$$E \left[(f(x) - E[f(x)] + E[f(x)] - \hat{f}(x))^2 \right] = E \left[(f(x) - E[f(x)])^2 \right] + E \left[(E[f(x)] - \hat{f}(x))^2 \right] + 2 E \left[(f(x) - E[f(x)])(E[f(x)] - \hat{f}(x)) \right]$$

>> Section 6

In classification The bias–variance decomposition was originally formulated for least-squares regression. For the case of classification under the 0-1 loss (misclassification rate), it is possible to find a similar decomposition, with the caveat that the variance term becomes dependent on the target label. Alternatively, if the classification problem can be phrased as probabilistic classification, then the expected cross-entropy can instead be decomposed to give bias and variance terms with the same semantics but taking a different form. It has been argued that as training data increases, the variance of learned models will tend to decrease, and hence that as training data quantity increases, error is minimised by methods that learn models with lesser bias, and that conversely, for smaller training data quantities it is ever more important to minimise variance.

>> Section 7

$$\text{MSE} = E \left[(y - \hat{f}(x; D))^2 \right] = (\text{Bias } D)^2 + \text{Var } D + \sigma^2$$

>> Section 8

the square of the bias of the learning method, which can be thought of as the error caused by the simplifying assumptions built into the method. E.g., when

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approximating a non-linear function $f(x)$ using a learning method for linear models, there will be error in the estimates $\hat{f}(x)$ due to this assumption; the variance of the learning method, or, intuitively, how much the learning method $\hat{f}(x)$ will move around its mean; the irreducible error σ^2 . Since all three terms are non-negative, the irreducible error forms a lower bound on the expected error on unseen samples. The more complex the model $\hat{f}(x)$ is, the more data points it will capture, and the lower the bias will be. However, complexity will make the model "move" more to capture the data points, and hence its variance will be larger.

>> Section 9

The bias error is an error from erroneous assumptions in the learning algorithm. High bias can cause an algorithm to miss the relevant relations between features and target outputs (underfitting). The variance is an error from sensitivity to small fluctuations in the training set. High variance may result from an algorithm modeling the random noise in the training data (overfitting). The bias–variance decomposition is a way of analyzing a learning algorithm's expected generalization error with respect to a particular problem as a sum of three terms, the bias, variance, and a quantity called the irreducible error, resulting from noise in the problem itself.